

Quantum Machine Learning for Calorimeter Simulation

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Topic: DMA

Generative Modeling for Calorimeter Data

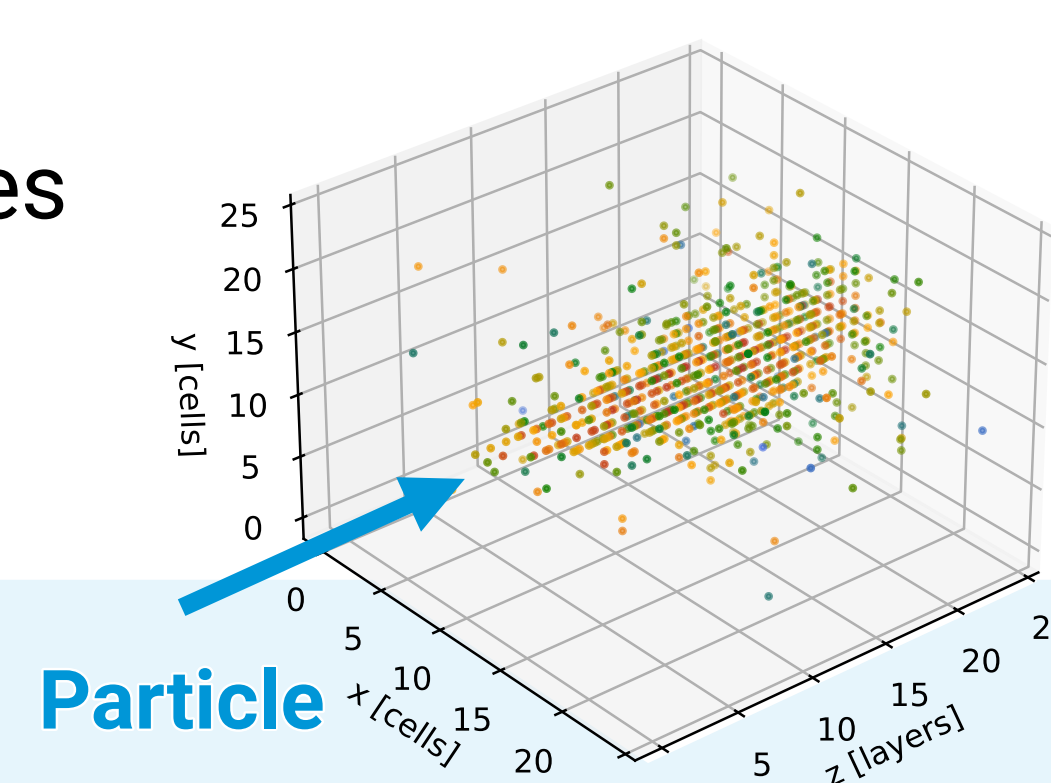
Use case: Simulation of shower images from calorimeters with QML.

Dataset: Shower images recorded by particles within the energy range of [225, 275] GeV.

Potential advantages with QML:

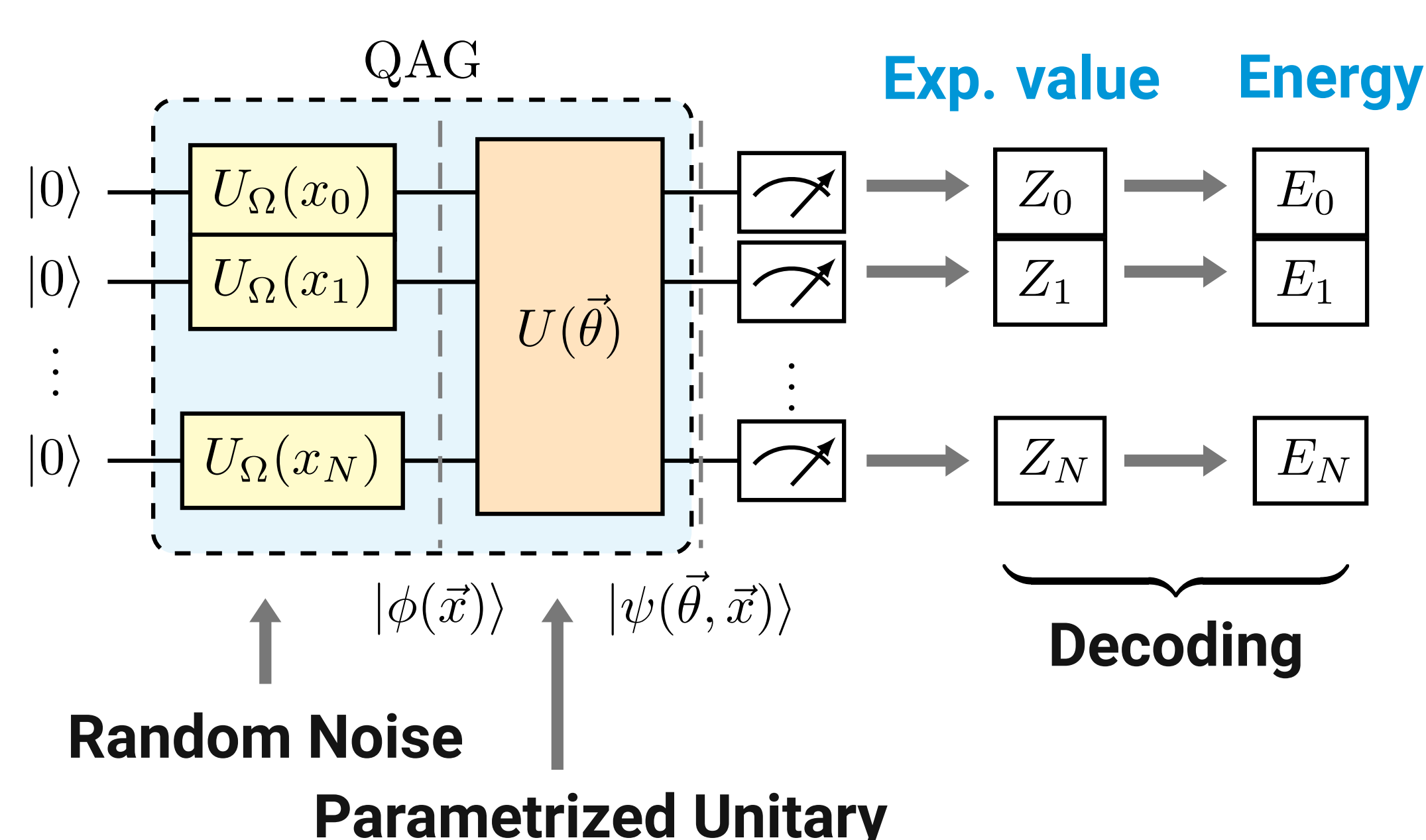
- Better generalization capabilities
- Fewer parameters needed
- Higher expressivity

Classical approach: See talk from F. Gaede "Generative ML for fast Detector Simulation".

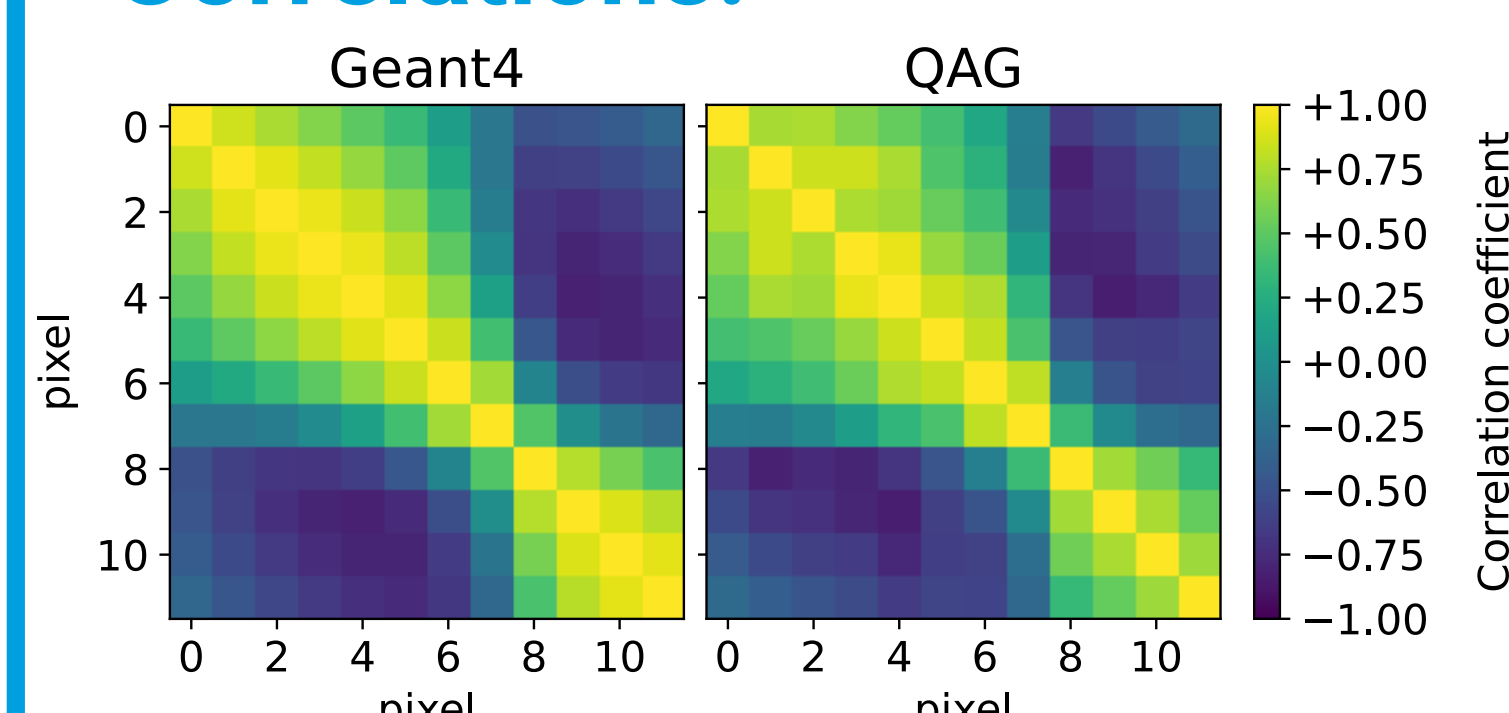


Quantum Angle Generator

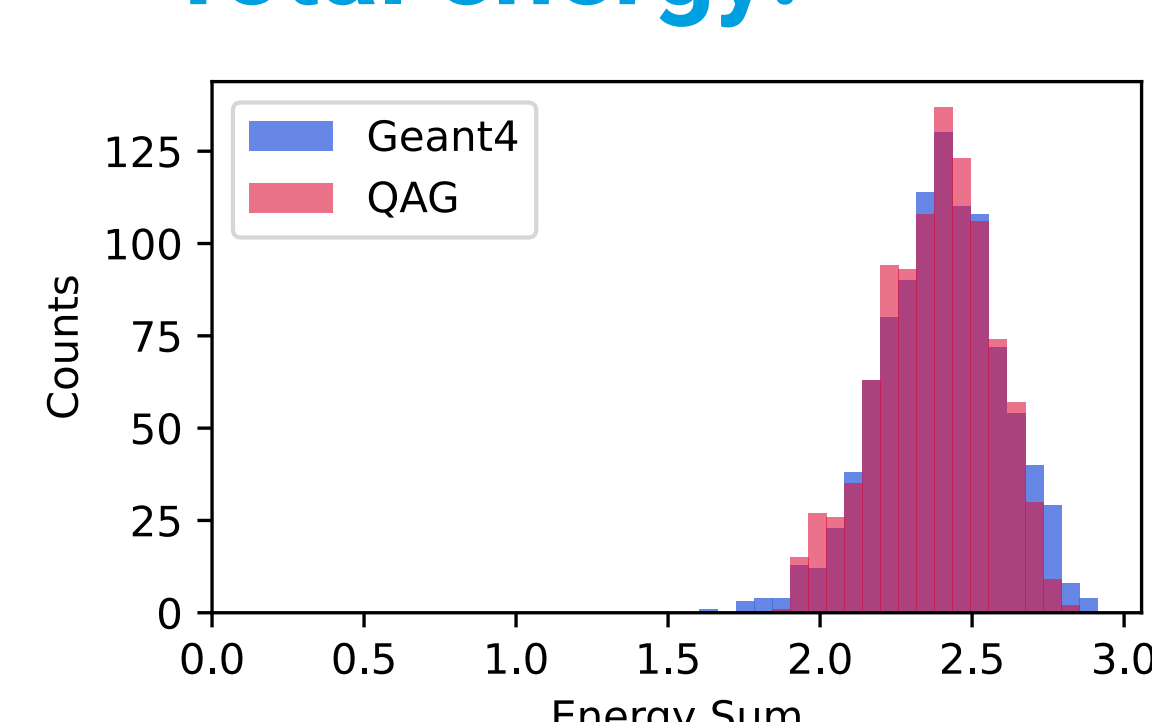
- **Encoding:** Classical pixel energy into a quantum state, in form of a rotational angle of a qubit.
- **Decoding:** Translate quantum state into an energy by measuring qubit multiple times (n_{shots}).



Correlations:

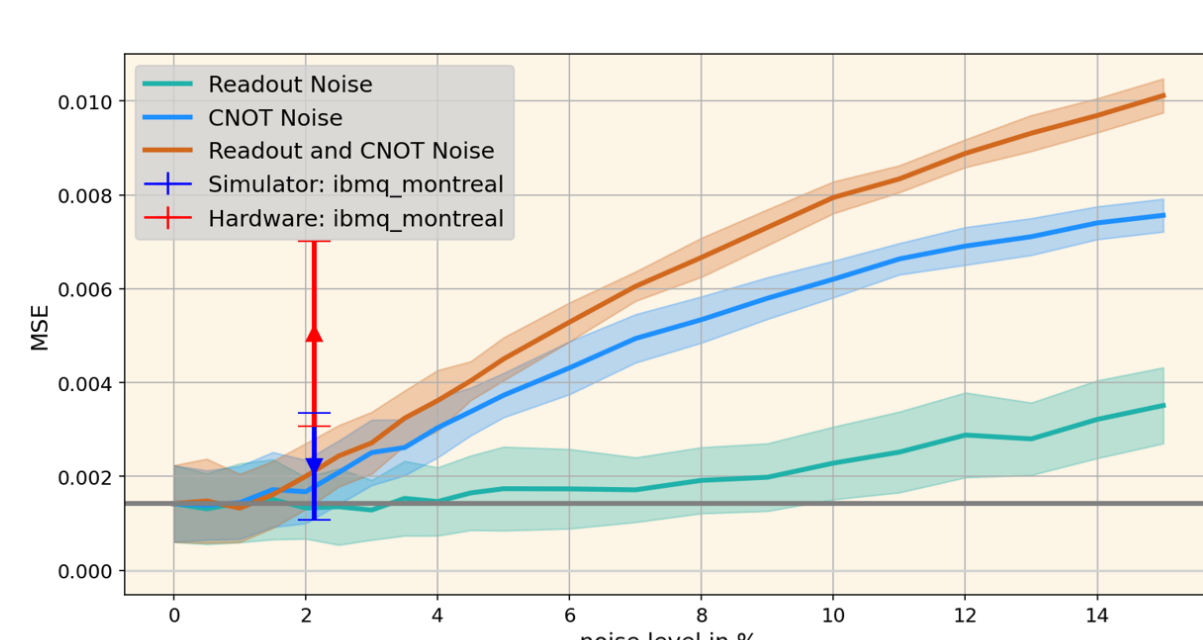


Total energy:

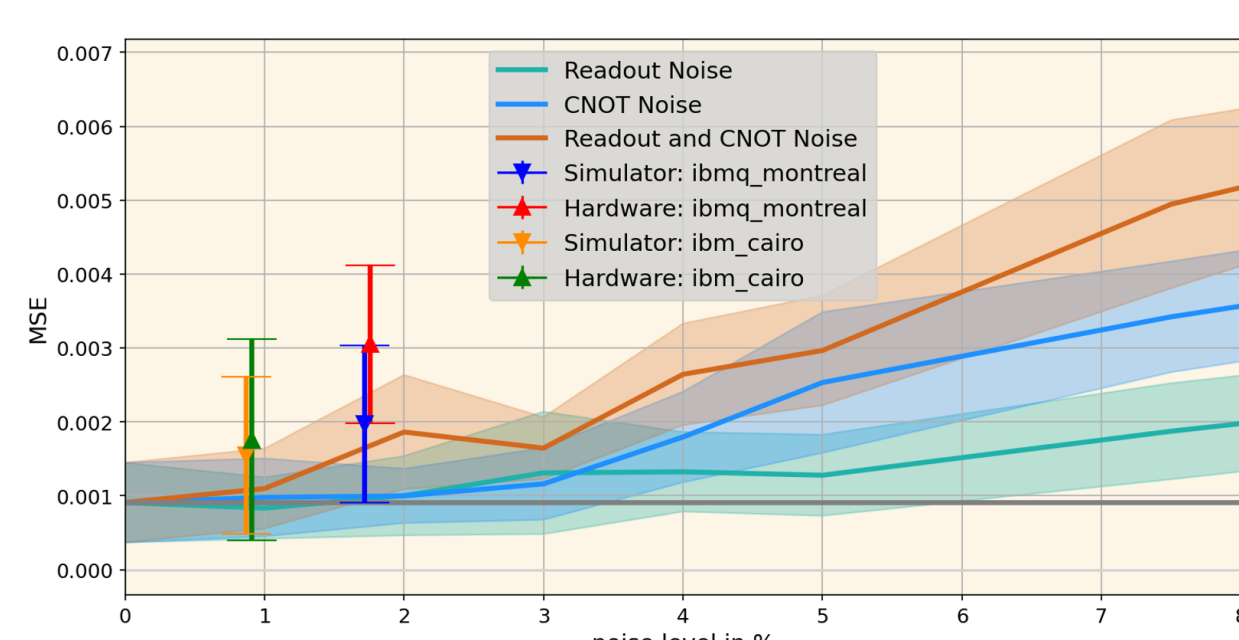


Noise study:

Inference:
Noise only during inference



Training:
Noise during training phase



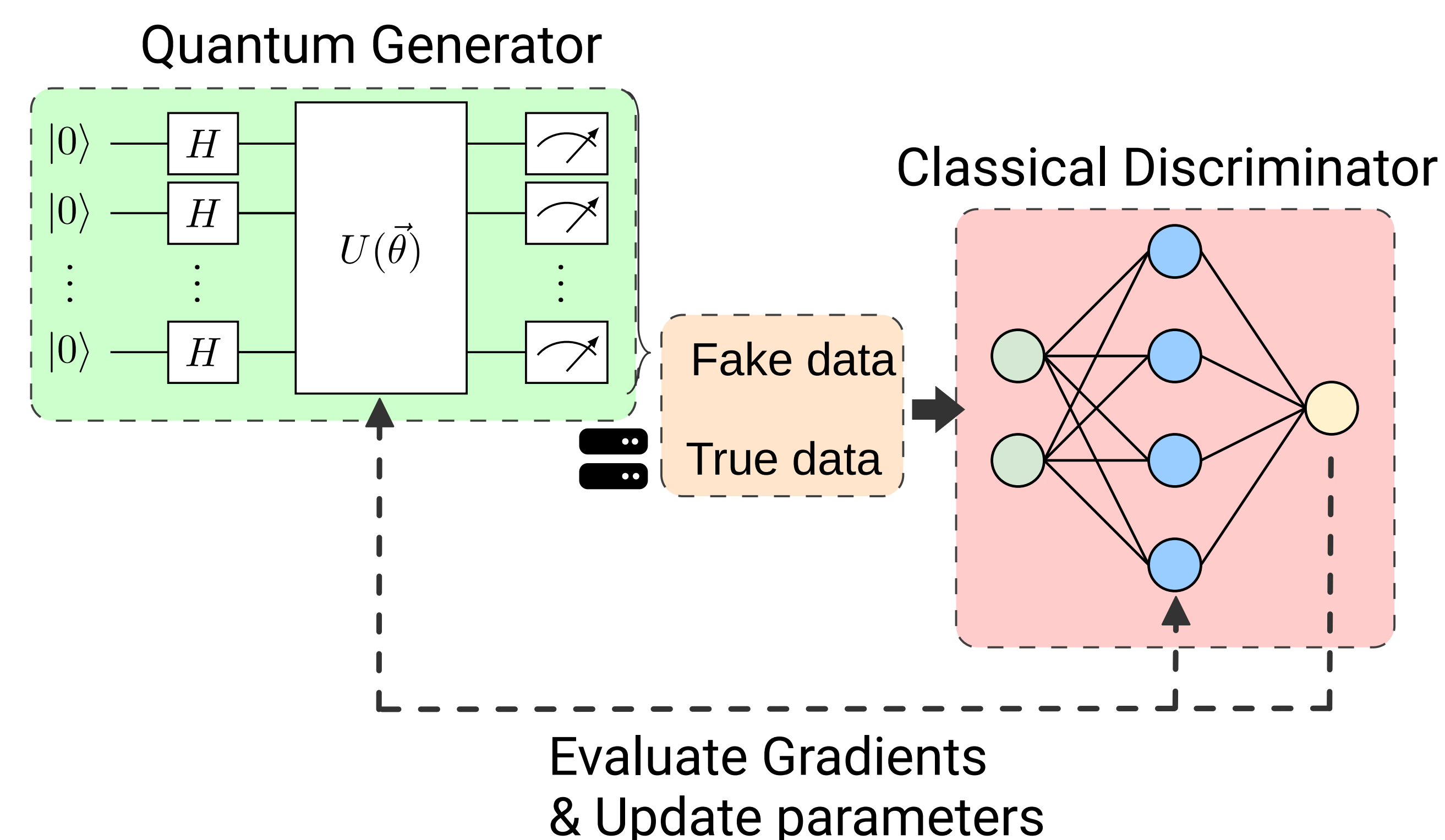
- ★ Smooth training with fast convergence
- ★ 20x less parameters compared to classical counterparts
- ★ Adaptation to the underlying noise

References

- [1] Florian Rehm: *Deep learning and quantum generative models for high energy physics calorimeter simulations*, RWTH Aachen University, PhD Dissertation (2023)
- [2] Florian Rehm, Sofia Vallecorsa, Kerstin Borrás, Dirk Krücker, Michele Grossi, Valle Varo: *Precise Image Generation on Current Noisy Quantum Computing Devices*, Quantum Science and Technology (2023)

Quantum Generative Adversarial Network

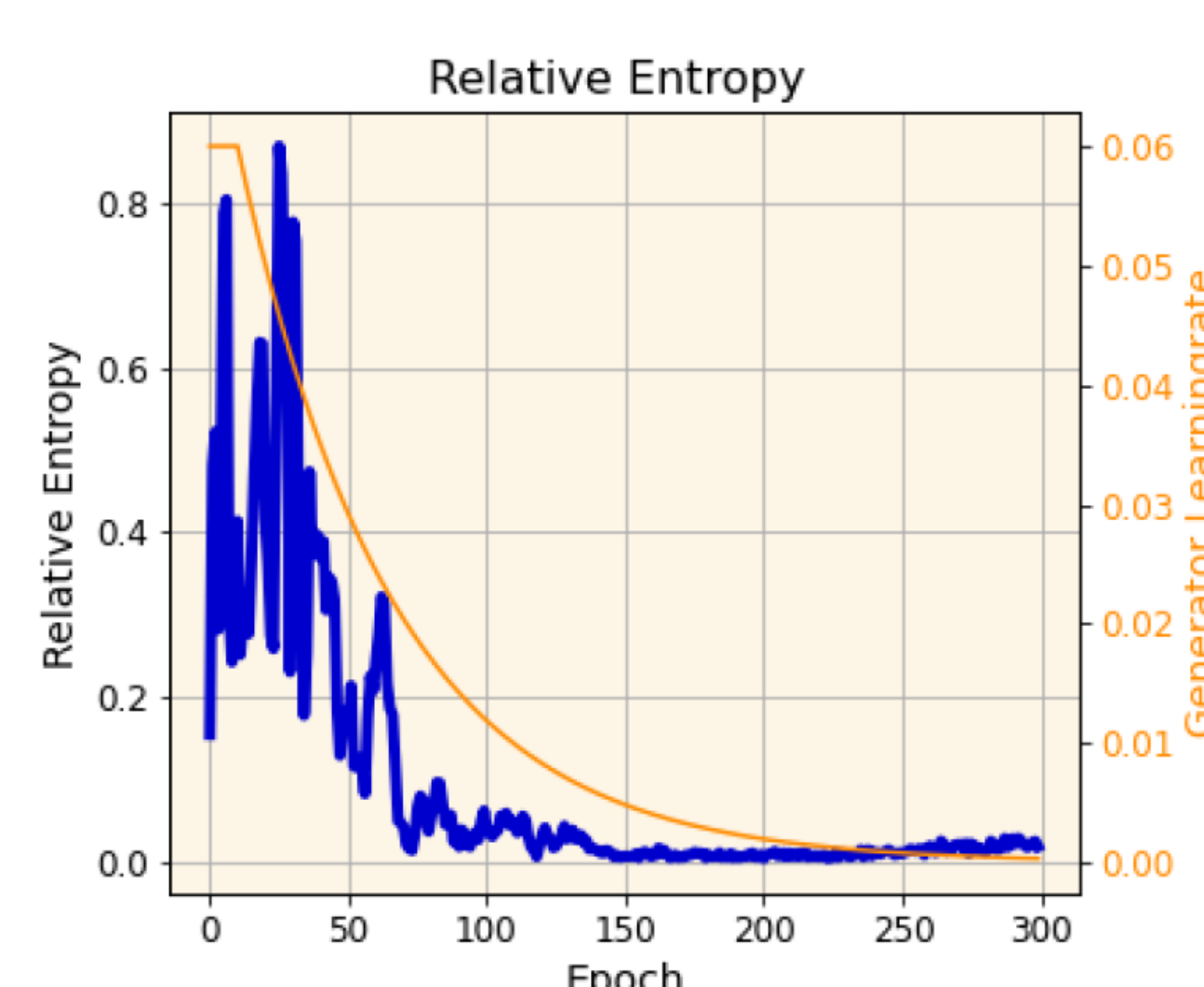
- **Hybrid Design:** Quantum Generator + Classical Discriminator



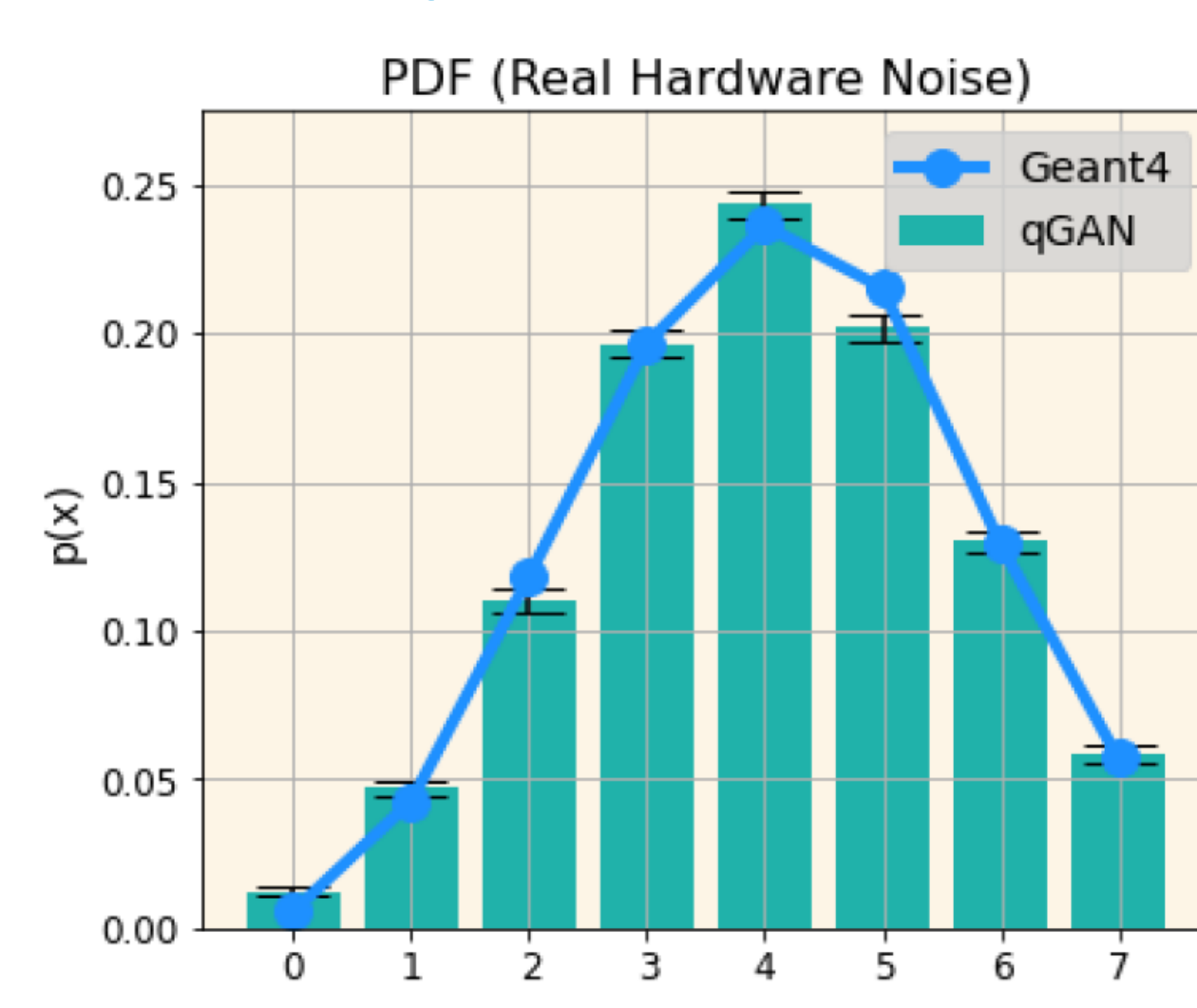
Performances

Generator run on *ibmq athens* quantum device.

Training:



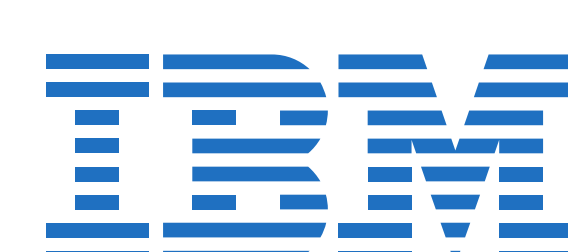
Energy PDF:



★ High scalability through amplitude encoding

Implementation on real hardware

Test of the architecture on real quantum processors under different noise type and levels.



IBM quantum computers:

- Type: Superconducting
- Processors: *Osprey*, *Condor*



IQM quantum computers:

- **Deneb:** Superconducting QC with resonator
- **Garnet:** Superconducting QC with grid connectivity



eleQtron quantum computers:
Ion-trap quantum computers

About me

Academic & Research Path

- **PhD** in Quantum Generative models for High Energy Physics - ENGAGE Marie Skłodowska-Curie PhD (2024-Current)
- **M.Sc. Thesis** in Jet Tagging with Quantum Classifiers (2021-2023)
- **Internship @ CERN** for phase detection of spin systems with Supervised and Unsupervised QML models (2022)

👤 **Contact**

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